
EMIC and IWYMIC Performance as a Signal for Later Olympiad Success

APMO and IMO Outcomes Through 2026

Full Statistical Analysis Report

Fixed identity roster: 3,112 contestants
EMIC/IWYMIC study years: 2013 to 2023, excluding 2020
Higher-contest endpoint: 2026
Generated: 2026-07-23

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1 Executive Summary

This report asks whether stronger performance at a contestant’s first observed EMIC or IWYMIC award appearance predicts later APMO or IMO success. The analysis finds a consistent predictive association among named EMIC/IWYMIC award recipients. It does not estimate a causal effect of participation.

Main result. Adding baseline percentile to a country, contest-stage, and baseline-year model improves out-of-time average precision for all four outcomes. The improvement is 0.088 for APMO awards and 0.075 for IMO awards. Both country-cluster bootstrap intervals remain above zero.

- In unrestricted follow-up through 2026, the APMO award rate rises from 4.4% in the 25-50% baseline band to 17.7% in the 75-100% band. The IMO rate rises from 5.8% to 17.6%.
- At five years, the right-censored award incidence for the 75-100% band is 16.6% for APMO and 17.2% for IMO. The corresponding 25-50% rates are 4.9% and 5.6%.
- Baseline award tier matters most clearly at IWYMIC. Within five years, 30.8% of IWYMIC Gold recipients earn an APMO award and 36.6% earn an IMO award.
- Country remains important. Rates differ sharply across delegations, and many country intervals are wide. Performance is informative, but selection systems and training environments also shape outcomes.

2 Data Scope and 2026 Update

The identity universe is fixed by EMIC and IWYMIC. APMO and IMO records can enrich an existing identity but cannot add a new person. This design keeps the research question centered on former EMIC/IWYMIC contestants.

Table 1: Contest Coverage in the Fixed Roster

Contest	Roster Identities	Matched Appearances	Coverage
EMIC	1,814	2,008	2013 to 2023, excluding 2020
IWYMIC	1,646	1,900	2013 to 2023, excluding 2020
APMO	360	634	2016 to 2026
IMO	375	742	2013 to 2026

The official 2026 APMO report contains 397 contestants, of whom 55 match the fixed roster. The official 2026 IMO page contains 666 contestants, of whom 55 match the roster. All IMO official ranks reconcile with recomputed global score ties. All APMO award labels reconcile with the official cutoffs and country award limits.

APMO source status. The APMO 2026 score-level report is complete but the official site still labels it preliminary. The report is included because the user requested the current 2026 data. Its status is retained in the audit and changelog. A future refresh should be run after the official preliminary notice is removed.

APMO source anomaly. The APMO 2026 Vietnam rank-2 row displays problem scores totaling 18 and an official total of 25. The official total is retained for rank and award calculations. This is the only 2026 problem-total mismatch.

Official sources: https://www.apmo-official.org/year_report/2026 and <https://www.imo-official.org/results/individual/year/2026/>.

3 Definitions and Methods

3.1 Baseline and Outcomes

- The unit of analysis is one reviewed contestant identity.
- Baseline is the earliest EMIC or IWYMIC award appearance. EMIC breaks a same-year tie.
- Baseline rank uses the average position of the full score tie. Percentile is $1 - \text{rank average}/\text{participants}$.
- A later award is Gold, Silver, Bronze, or Honourable Mention. A later medal is Gold, Silver, or Bronze.
- APMO and IMO outcomes must occur in a strictly later calendar year. Same-year and earlier results are excluded.
- The primary prediction window is five calendar years. Complete-window eligibility requires all five years to be observable through 2026.

The complete five-year samples contain 1,754 contestants for APMO and 2,486 for IMO. APMO baselines before 2015 are excluded from complete-window analysis because complete contestant-level APMO data begin in 2016.

3.2 Analytical Components

1. Descriptive progression rates compare fixed baseline-percentile bands.
2. Raw quintiles and adjusted curves assess the continuous performance signal.
3. Outcome transitions connect baseline award tier to later result class.
4. Kaplan-Meier-style cumulative incidence handles right-censoring.
5. Country estimates use Wilson 95% intervals.
6. Ridge-logistic models use forward baseline-year validation. Each test year is predicted only from earlier baseline years.
7. The baseline model uses country, stage, and baseline year. The performance model adds percentile, percentile squared, and a percentile-by-stage interaction.
8. Metric-difference intervals use 300 country-cluster bootstrap replicates with a fixed seed.

4 Observed Progression by Baseline Performance

The unrestricted descriptive rates include every later record observed through 2026. They show the full observed progression but give older cohorts more follow-up time than recent cohorts.

Table 2: Unrestricted Progression Through 2026

Baseline Band	Contestants	APMO Award	APMO Medal	IMO Award	IMO Medal
25-50%	936	4.4%	2.9%	5.8%	3.8%
50-75%	1,136	9.4%	6.4%	10.0%	6.9%
75-100%	1,040	17.7%	14.1%	17.6%	14.9%

The top band has about 4.0 times the APMO award rate and 3.1 times the IMO award rate of the 25-50% band. These ratios are descriptive. The fixed-window models below provide the stronger predictive test.

Five-Year Higher-Contest Award Probability Rises With Baseline Performance

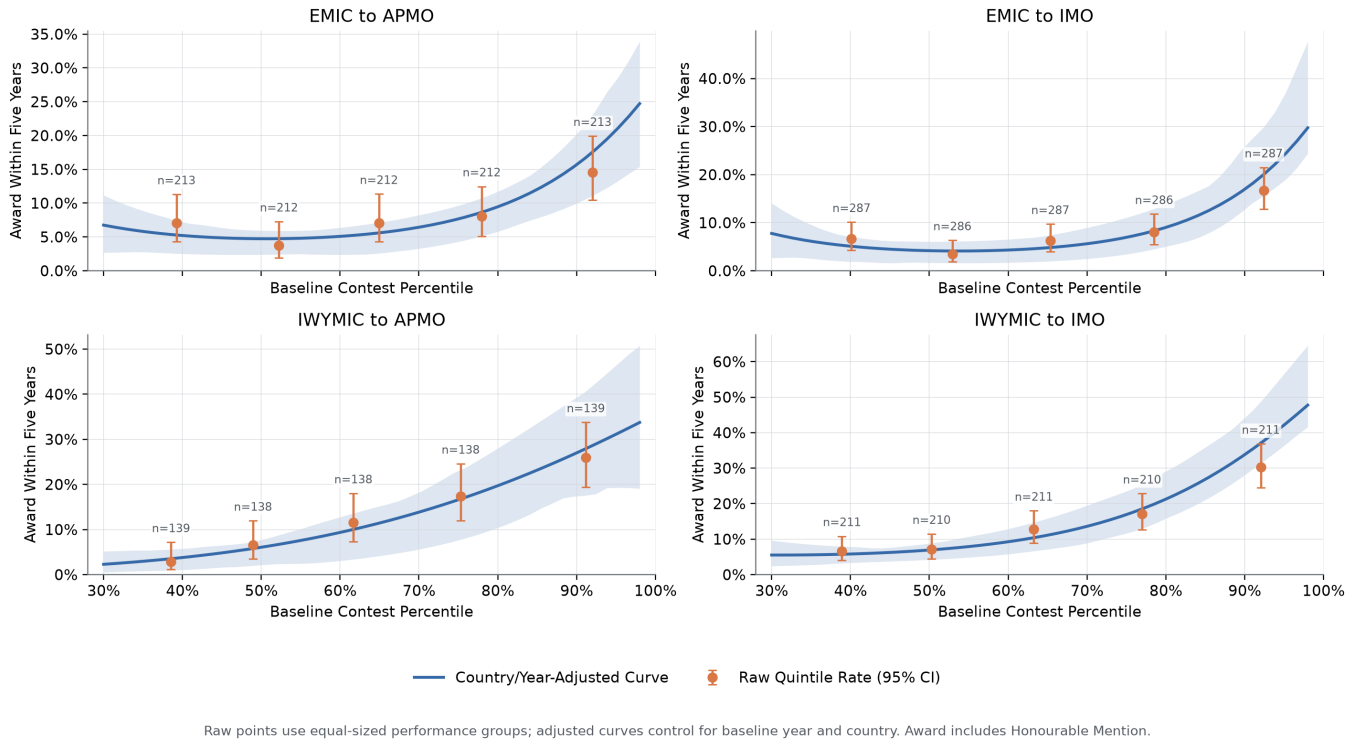


Figure 1: Raw five-year quintile rates and country/year-adjusted curves. Labels show quintile sample sizes.

The adjusted curves rise most sharply near the upper end of the observed percentile range. IWYMIC performance has the steepest relationship with both APMO and IMO outcomes. The uncertainty bands widen near the extremes because fewer observations support those curve regions.

5 Baseline Award Tier and Later Result Class

Baseline medal tiers provide an ordinal sensitivity check that does not depend on the precision of the total-participant denominator. Higher baseline tiers generally lead to higher five-year award rates.

Table 3: Five-Year Higher-Contest Award Rate by Baseline Tier

Stage	Baseline Tier	APMO Award	IMO Award
EMIC	Merit	5.9%	5.1%
EMIC	Bronze	6.8%	7.0%
EMIC	Silver	10.3%	8.9%
EMIC	Gold	18.6%	24.8%
IWYMIC	Merit	4.9%	7.1%
IWYMIC	Bronze	15.2%	14.7%
IWYMIC	Silver	21.9%	23.2%
IWYMIC	Gold	30.8%	36.6%

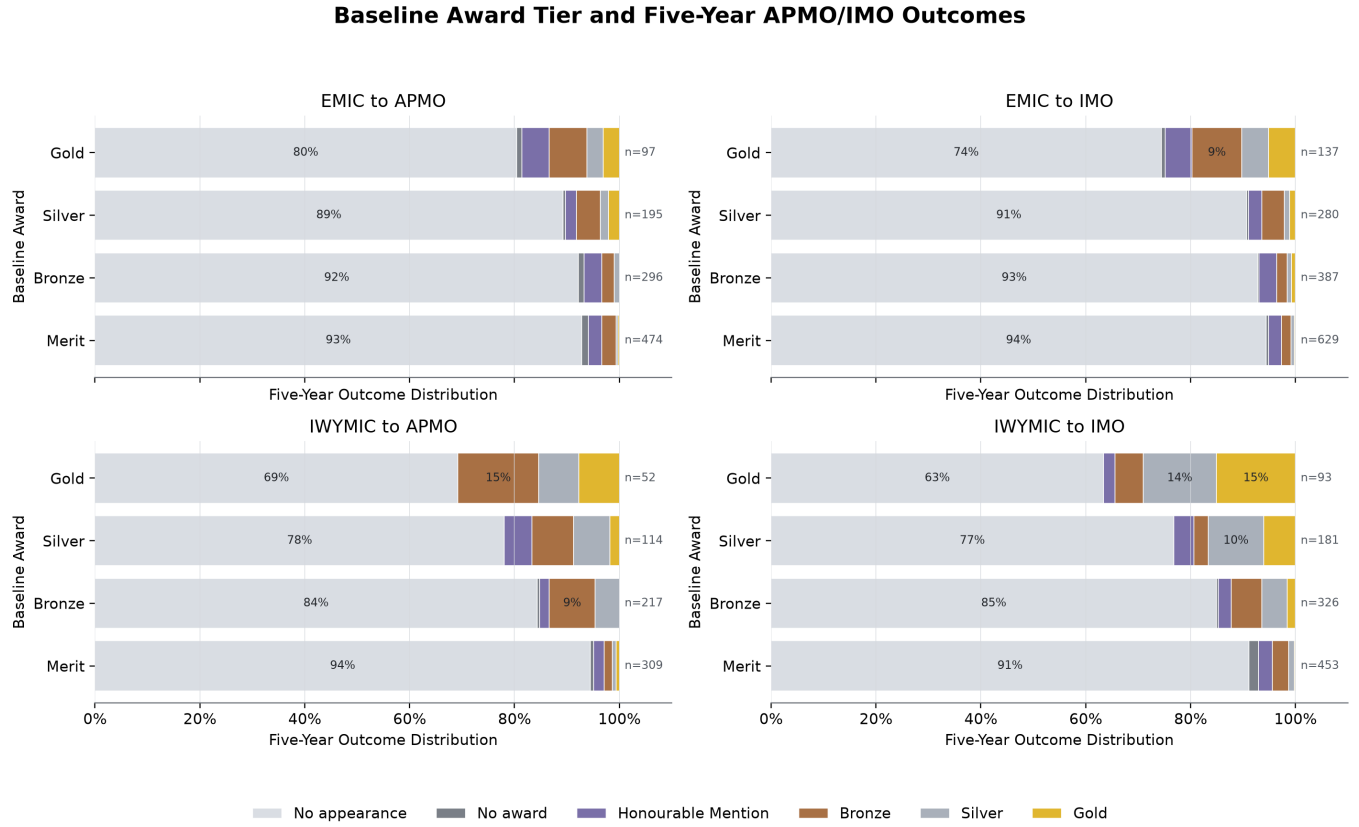


Figure 2: Distribution of later outcomes within five years, grouped by first observed stage and award tier.

The largest separation appears among IWYMIC recipients. IWYMIC Gold recipients have five-year award rates of 30.8% for APMO and 36.6% for IMO, compared with 4.9% and 7.1% for IWYMIC Merit recipients.

6 Timing of Later Awards

The time-to-event analysis retains incomplete recent cohorts by censoring each person at the end of 2026. It estimates the cumulative probability of receiving a later award among contestants whose outcome window begins within source coverage.

Table 4: Five-Year Cumulative Award Incidence

Destination	Baseline Band	Group	Estimate	Lower 95%	Upper 95%
APMO	25-50%	746	4.9%	3.3%	6.6%
APMO	50-75%	872	10.2%	8.1%	12.3%
APMO	75-100%	762	16.6%	13.8%	19.4%
IMO	25-50%	936	5.6%	4.0%	7.1%
IMO	50-75%	1,136	9.4%	7.6%	11.1%
IMO	75-100%	1,040	17.2%	14.8%	19.5%

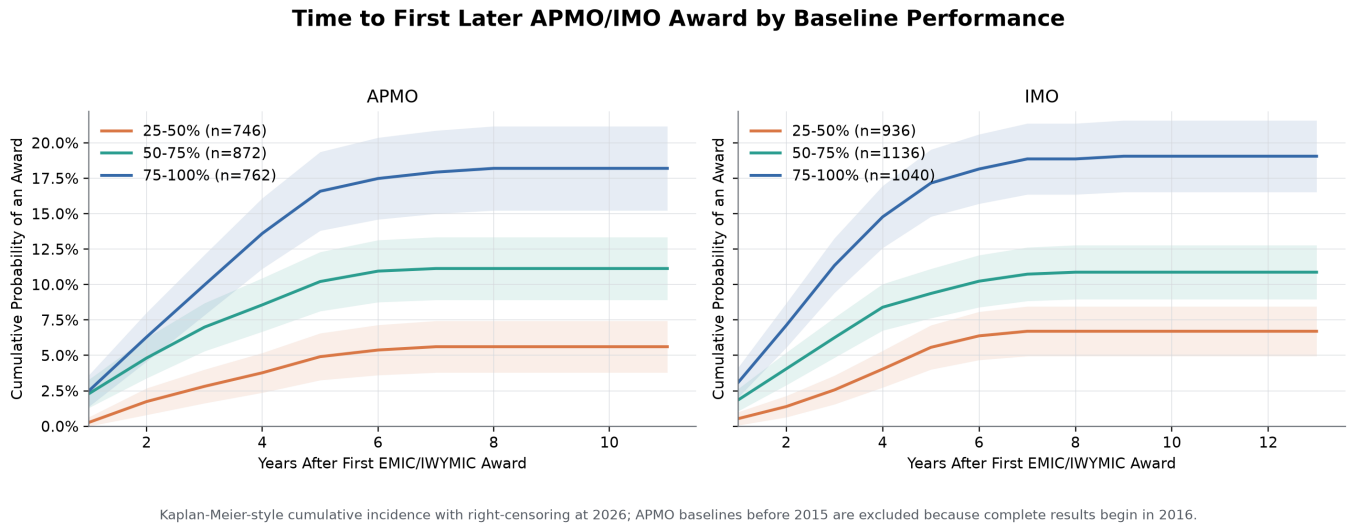


Figure 3: Cumulative probability of a first later award with right-censoring at 2026.

Most separation develops within the first five to seven years after baseline. The curves then flatten because few new awards occur at longer lags and fewer contestants remain under observation.

7 Country Context

Country rates reflect both contestant performance and delegation systems. They should not be read as isolated measures of national program quality. Small samples can produce high point estimates with wide intervals.

Table 5: Highest Five-Year Award Rates Among Countries With at Least 30 Eligible Contestants

Destination	Country	Eligible	Awards	Rate	Wilson 95% Interval
APMO	Tajikistan	44	23	52.3%	37.9% to 66.2%
APMO	Mexico	47	22	46.8%	33.3% to 60.8%
APMO	Sri Lanka	31	8	25.8%	13.7% to 43.2%
APMO	Kazakhstan	46	8	17.4%	9.1% to 30.7%
APMO	Malaysia	43	7	16.3%	8.1% to 30.0%
IMO	Uzbekistan	33	16	48.5%	32.5% to 64.8%
IMO	Sri Lanka	51	20	39.2%	27.0% to 52.9%
IMO	Tajikistan	44	17	38.6%	25.7% to 53.4%
IMO	Singapore	34	13	38.2%	23.9% to 55.0%
IMO	Mexico	59	19	32.2%	21.7% to 44.9%

Country Progression Rates Require Uncertainty and Denominator Context

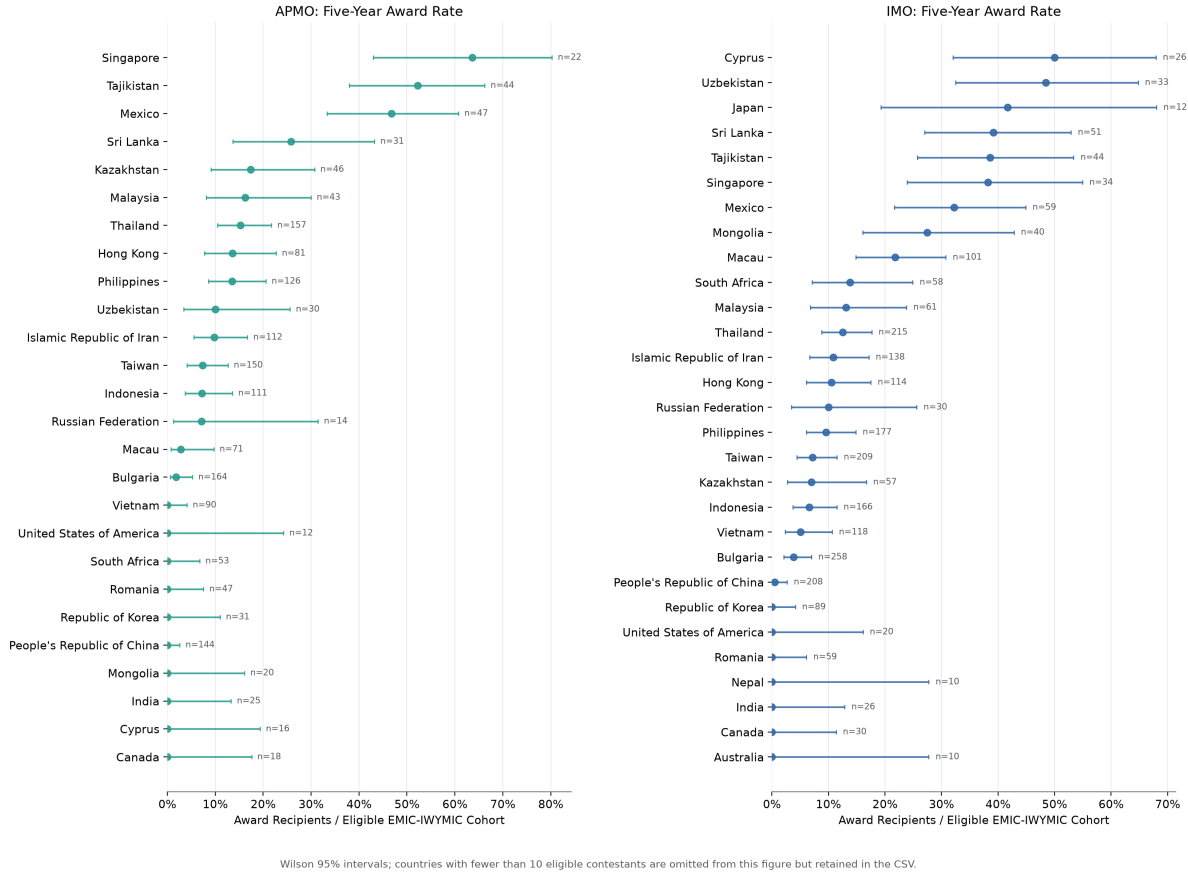


Figure 4: Five-year country award rates. The figure shows countries with at least 10 eligible contestants; the table above applies a stricter threshold of 30.

The large country differences justify including country in the baseline model. They also show why raw cross-country comparisons can be misleading without denominator and uncertainty information.

8 Out-of-Time Predictive Models

Forward validation tests whether the model works on later cohorts. APMO predictions cover baseline years 2018, 2019, and 2021. IMO predictions cover 2016, 2017, 2018, 2019, and 2021. No model is trained on its test year or any later year.

Table 6: Out-of-Time Predictive Performance

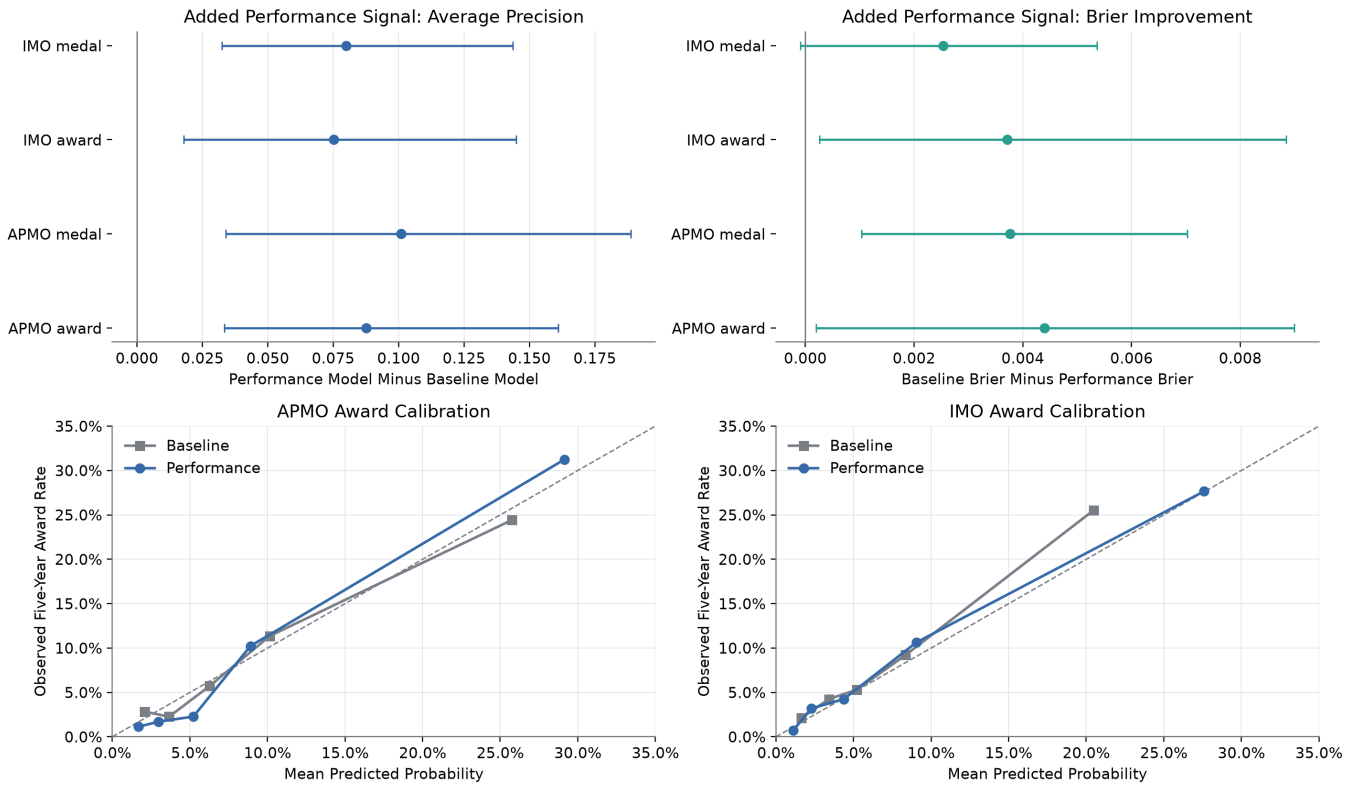
Destination	Outcome	Model	Predictions	Events	Avg. Precision	ROC AUC	Brier
APMO	Award	Baseline	879	82	0.278	0.767	0.076
APMO	Award	Performance	879	82	0.365	0.838	0.071
APMO	Medal	Baseline	879	59	0.221	0.767	0.058
APMO	Medal	Performance	879	59	0.322	0.848	0.054
IMO	Award	Baseline	1,411	131	0.276	0.763	0.076
IMO	Award	Performance	1,411	131	0.351	0.809	0.072
IMO	Medal	Baseline	1,411	91	0.153	0.712	0.058
IMO	Medal	Performance	1,411	91	0.233	0.797	0.055

Table 7: Incremental Value of Adding Baseline Performance

Destination	Outcome	Δ Avg. Precision	95% Interval	Δ ROC AUC	Brier Improvement	95% Interval
APMO	Award	0.088	0.033 to 0.161	0.071	0.004	0.000 to 0.009
APMO	Medal	0.101	0.034 to 0.189	0.081	0.004	0.001 to 0.007
IMO	Award	0.075	0.018 to 0.145	0.046	0.004	0.000 to 0.009
IMO	Medal	0.080	0.032 to 0.144	0.085	0.003	0.000 to 0.005

Average precision improves by 0.088 for APMO awards, 0.101 for APMO medals, 0.075 for IMO awards, and 0.080 for IMO medals. Every average-precision interval is above zero. Brier scores improve at the point estimate for all outcomes, although the IMO medal interval slightly crosses zero.

Out-of-Time Predictive Value and Calibration



Baseline model: country, stage, and baseline year. Performance model adds baseline percentile, curvature, and a stage interaction. Intervals use country-cluster bootstrap resampling.

Figure 5: Metric improvement and calibration for forward-year predictions. Positive improvement favors the performance model.

The performance model improves ranking and discrimination. Calibration remains broadly reasonable, but the highest-risk bins contain fewer events and therefore show more variation. The model is useful for group-level signal assessment. It is not suitable for deterministic predictions about individual students.

9 Cohort Coverage

Coverage determines which observations can support a fixed five-year outcome. Baselines from 2021 now have a complete five-year window because the higher-contest endpoint is 2026. Baselines from 2022 and

2023 remain incomplete and are excluded from fixed-window models.

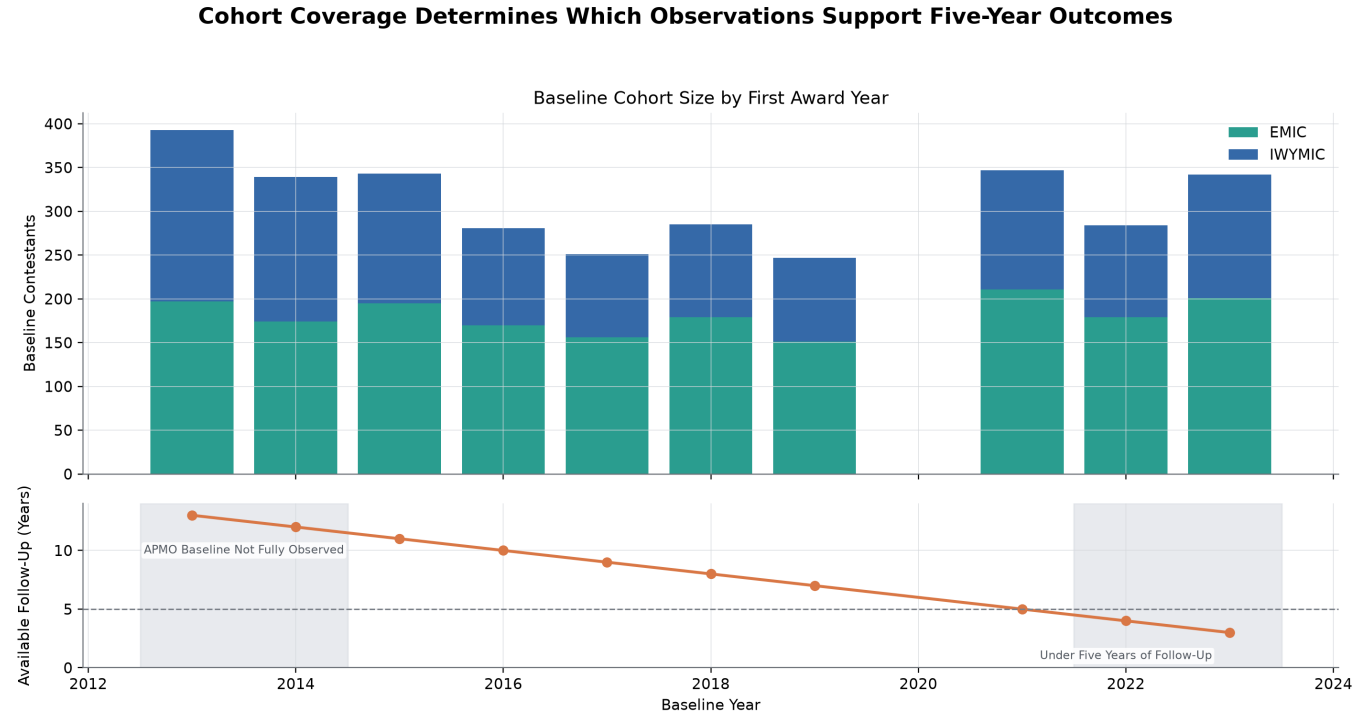


Figure 6: Baseline cohort sizes and available follow-up through 2026. Shading marks incomplete APMO source coverage and baselines with fewer than five follow-up years.

The 2026 update increases the five-year eligible samples to 1,754 for APMO and 2,486 for IMO. This is a substantive gain over the 2025 endpoint because an additional baseline cohort enters both model evaluation and descriptive fixed-window analyses.

10 Answer to the Research Question

Conclusion. Stronger EMIC/IWYMIC performance is a useful signal for later APMO and IMO success among named award recipients. The signal remains after adjustment for country, baseline year, and stage, and it improves prediction on later cohorts.

The evidence does not show that EMIC or IWYMIC participation causes later success. Strong contestants may already have better preparation, training access, selection opportunities, or prior mathematical development. The appropriate interpretation is predictive association.

The evidence is strongest when several views agree:

- Percentile curves rise with baseline performance.
- Medal-tier transitions show the same ordering.
- Time-to-event estimates separate early and remain ordered.
- Out-of-time models improve when performance is added.
- Country adjustment reduces, but does not remove, the signal.

11 Limitations

- EMIC/IWYMIC result pages name award recipients only. Unnamed non-awardees cannot be followed, so the study does not compare all participants.
- Country selection, training access, age, and prior preparation are not observed.
- IWYMIC total-participant denominators are estimated. Medal-tier results provide an important sensitivity check.
- APMO 2013 to 2015 lacks complete contestant-level score tables and is excluded from complete-window outcomes.
- APMO 2026 is preliminary and may change when finalized.
- Recent cohorts are right-censored. Fixed-window models intentionally exclude incomplete follow-up.
- Identity matching is conservative. Ambiguous common names remain unmatched, which can slightly understate progression.

12 Reproducibility and Output Inventory

Run the statistical pipeline and report builder from the project root:

```
python "2 Processing Scripts/analyze_performance_signal.py"
python "2 Processing Scripts/build_statistical_analysis_report.py" --compile
```

Table 8: Analysis Tables Used in This Report

File	Role
Analysis Cohort.csv	One derived row per fixed identity
Descriptive Progression Summary.csv	Observed later outcomes by fixed performance band
Performance Band Summary.csv	Raw five-year quintile rates and Wilson intervals
Adjusted Signal Curves.csv	Country/year-adjusted probability curves
Outcome Transition Summary.csv	Baseline award tier to later result class
Survival Summary.csv	Right-censored cumulative award incidence
Country Progression Intervals.csv	Country rates and Wilson intervals
Cohort Coverage.csv	Baseline-year coverage and follow-up eligibility
Model Predictions.csv	Forward-year out-of-time probabilities
Model Results.csv	Predictive metrics by model and outcome
Model Comparison.csv	Incremental metrics with country bootstrap intervals
Calibration Summary.csv	Predicted and observed rates by probability bin

The analysis code uses deterministic seeds for country-cluster bootstrap resampling. Generated tables, figures, the Markdown summary, this LaTeX source, the PDF, and source hashes are retained in **6 Statistical Analysis**.